GitHub Repo Live Dashboard

Explainable Machine Learning for Irish Residential Energy Retrofit Prioritisation Using BER Data

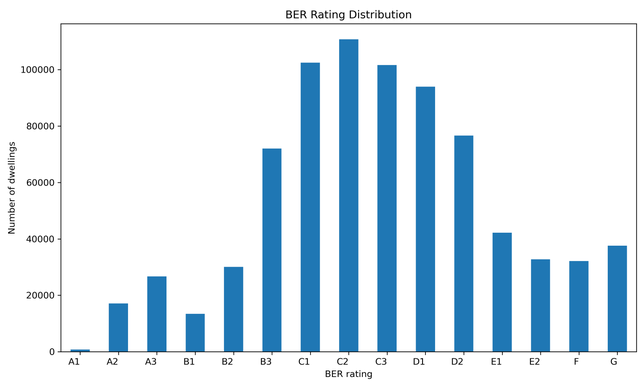
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ACM40960 – Projects in Mathematical Modelling | University College Dublin

Motivation & Research Question

- Residential retrofit is central to Ireland's climate targets, but hundreds of thousands of homes cannot be upgraded at once.
- **Can Building Energy Rating (BER) data predict which dwellings should be retrofit priorities – and can explainable ML reveal why?**
- Framed as decision support, not a replacement for a professional BER assessment.

Data



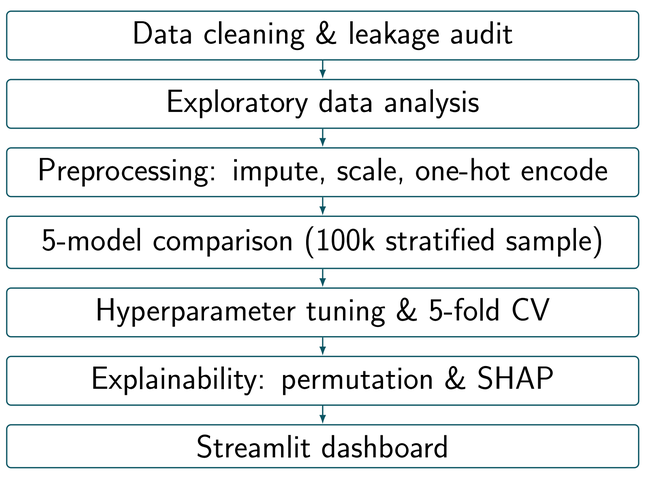
- Irish BER Database (cleaned), Ahern et al. (2024), Mendeley Data.
- **790,389** dwellings, 22 features after cleaning and a target-leakage audit (BER-derived output columns removed).
- Features: county, dwelling type, construction year, floor area, wall/roof/floor/window/door U-values and areas, heating fuels, glazing %, volume.

Target Definition – Why B2?

Label	Condition
Not priority (0)	BER **B2** or better
Retrofit priority (1)	Worse than B2

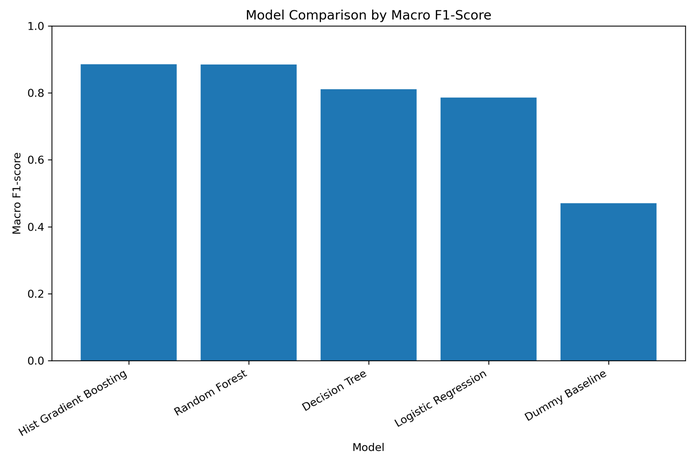
- Class balance: **88.8% priority** vs 11.2% not priority (imbalanced → balanced accuracy & macro F1 used throughout, not raw accuracy).
- **B2 is not arbitrary**: it is SEAI's own recommended minimum retrofit standard, and the explicit target of the National Retrofit Plan – **500,000 homes to B2 by 2030**.
- A model-flagged "priority" dwelling is, by definition, one that has not yet met the national retrofit benchmark.

Methodology



Model Comparison

Five models compared on a stratified 100k-row sample (80k train / 20k test): a majority-class dummy baseline, Logistic Regression, Decision Tree, Random Forest, and Hist Gradient Boosting.



- The dummy baseline reaches 88.8% *accuracy* by always predicting the majority class – but scores **0.50 balanced accuracy and 0.50 ROC-AUC**: no real discrimination ability.
- **Hist Gradient Boosting** gave the strongest overall trade-off (macro F1, ROC-AUC, high priority recall) and was selected as the final model.

Tuning & Validation

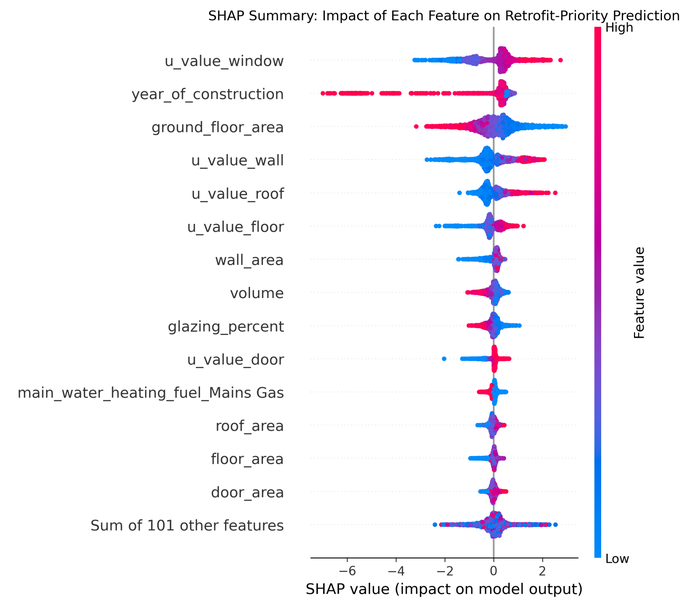
- **RandomizedSearchCV**: 25 candidates × 5-fold CV, scored on macro F1, over learning rate, iterations, leaf count, min. samples/leaf, and L2 regularisation.
- Tuning improved every held-out metric except priority recall (a negligible −0.0004 change).
- **5-fold stratified CV** on the training split confirms the single test-split result is not a lucky draw: macro F1 = 0.886 ± 0.005, closely matching the 0.892 test-set score.

	Acc.	Bal Acc	F1	AUC
Default	0.959	0.854	0.886	0.969
Tuned (final)	**0.960**	**0.863**	**0.892**	**0.970**

Does Sample Size Matter?

- Final model re-trained from scratch on 100k, 300k, and the **full 790,389-row** dataset (same tuned hyperparameters).
- Full-dataset training took under **40 seconds** – Hist Gradient Boosting scales well.
- Every metric improves modestly and monotonically with more data (macro F1: 0.892 → 0.904 at full scale) – confirming results are not an artefact of the smaller sample.

Explainability: SHAP & Permutation Importance



- Two independent methods agree closely: **8 of the top 10** features are shared between permutation importance and SHAP.
- Top drivers: **window, wall, and roof U-values**, **ground floor area**, **year of construction**, and **water-heating fuel** – all physically meaningful, consistent with building-energy principles.
- High (red) U-values push predictions toward retrofit priority; low (blue) U-values push toward not-priority – exactly the expected physical direction.
- SHAP also explains *individual* predictions in real units (e.g. actual U-values, m^2), not standardised model inputs – used live in the dashboard below.

Key Interpretation

BER data predicts retrofit priority with strong performance (**96% accuracy, 0.97 ROC-AUC**), and the features driving those predictions are exactly the ones building scientists would expect: envelope heat loss, construction age, and heating systems – not arbitrary statistical artefacts.

Interactive Dashboard

A Streamlit app puts the model in front of a user, not just a report:

- **Predict** – full input form for a dwelling's characteristics; returns the retrofit-priority prediction, probability, and a live SHAP waterfall explaining that specific prediction.
- **Model & Data** – comparison tables and key figures, doubling as a project showcase.
- **About** – project summary, data source, and limitations for a general audience.

Scan the QR codes above the title to try the live app or browse the full source code.

Limitations & Future Work

- Binary label simplifies a continuous energy measure at one cutoff – a D1 and a G-rated home are both "priority".
- BER reflects standardised assessment assumptions, not measured household energy use.
- Associations identified here are statistical, not causal.
- *Future:* multi-level thresholds, county/dwelling-type specific models, full-dataset production pipeline.

LaTeX TikZposter